



University
of Basel

High-order simulation of wave propagation using hybrid nonconforming methods on general meshes

Romain Mottier[§]

Joint work with: Alexandre Ern^{†‡}

Wave-Ned, Delft, August 2026

[§] Department of Mathematics and Computer Science, **University of Basel**, Switzerland

[†] CERMICS, **Ecole des national des Ponts et chaussées**, Marne la Vallée, France

[‡] SERENA Project-Team, **INRIA Paris**, France

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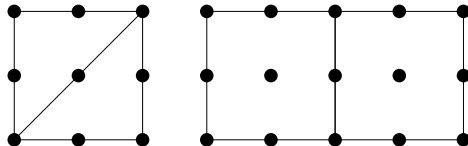
I.3 Further insight into HHO

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 - ✓ Allow for unstructured meshes
 - ✗ Non-local polynomial bases
 - ✗ Non-diagonal mass matrix:
Mass lumping required for explicit schemes



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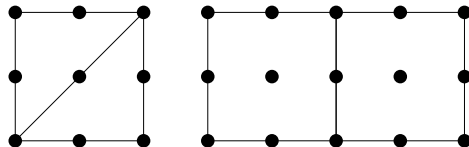
■ **Finite Differences (FD)**: Ideal for simple geometries 📄 [van Vossen, Robertsson, Chapman \(2002\)](#)

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✓ Allow for unstructured meshes

✗ Non-local polynomial bases

✗ Non-diagonal mass matrix:
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■ **Spectral Element Method (SEM)** 📄 [Patera \(1984\)](#) 📄 [Komatitsch, Vilotte \(1998\)](#) 📄 [Cohen \(2002\)](#)

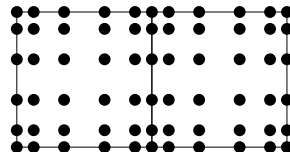
▶ **Current industrial standart**

▶ Quadrature points: tensorized GLL nodes

✓ **Diagonal mass matrix**

✓ Parallelizable and scalable for HPC applications

✗ **Mainly quadrangular/hexahedral meshes**



- **Big issue for wave problems in industry:** Meshing realistic geometries is time-costly (manual postprocessing)
 - ▶ Methods in industry rely mainly on quadrilateral/hexahedral meshes
 - ▶ Industrial meshers can generate meshes
 - with **different element shapes** (quadrangles+simplices / hexahedra+tetrahedra)
 - can include **hanging nodes**

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■ **Solution: Polytopal discretization methods**

QUADRANGLES



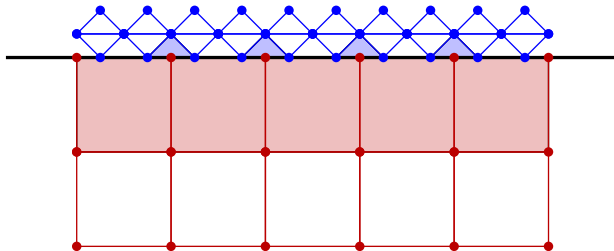
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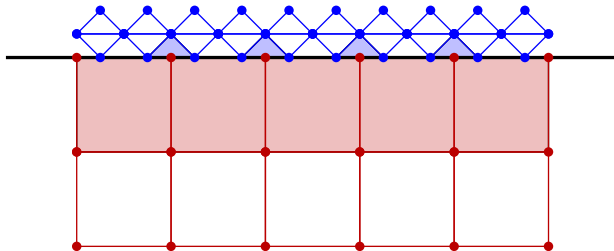
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■ **Two possibilities:**

- ▶ H^1 -conforming polytopal methods: **Virtual Element Methods (VEM)**
- ▶ Nonconforming methods: **class of discontinuous Galerkin methods**

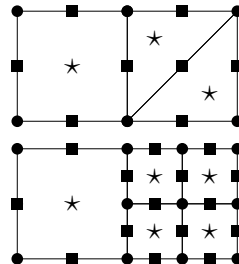
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📄 da Veiga, Brezzi, Cangiani, Manzini, Marini, Russo (2013)

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📄 Wriggers, Junker (2024)

✗ Same issue as cG: **Non-diagonal mass matrix**
(dofs attached to low-dimensional mesh entities)



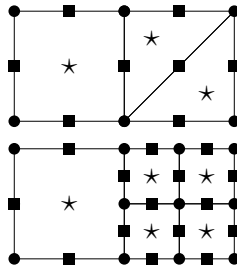
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✗ Same issue as cG: **Non-diagonal mass matrix**
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■ Discontinuous Galerkin methods (dG)

📄 Baker (1976) 📄 Wheeler (1978) 📄 Arnold (1982)

📄 Grote, Schneebeli, Schötzau (2006)

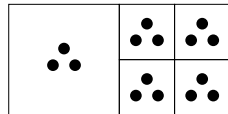
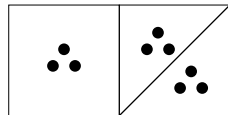
📄 Antonietti, Bonaldi, Mazzi (2020)

✓ Block-diagonal mass matrix

✗ More dofs than cG

✓ Compact stencil (HPC)

✗ **Stabilization:** Face-based
(Minimal value)



I.2. Hybrid Nonconforming Methods: HHO methods

- Strong links between the methods of this family (HHO, HDG, WG, ncVEM)



Cockburn, Di Pietro, Ern (2016)



Lemaire (2021)

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- Seminal papers:  Di Pietro, Ern, Lemaire (2014)  Di Pietro, Ern (2015)

- Wave propagation:  Burman, Duran, Ern (2022a)  Burman, Duran, Ern (2022b)
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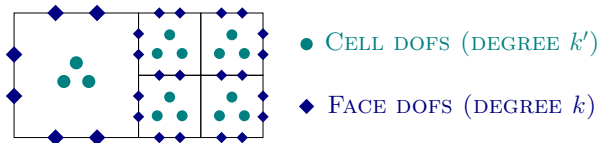
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- Idea: **Additional unknowns attached to mesh faces**

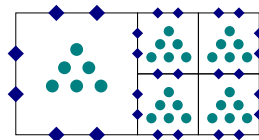
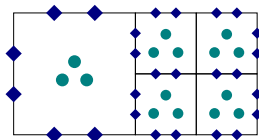


I.3. Further insight into HHO

- **Local degrees of freedom:** ▶ Cell unknowns: $u_T \in P^{k'}(T)$ ▶ Face unknowns: $u_{\partial T} \in P^k(\partial T)$

EQUAL-ORDER: $k' = k$ ($= 1$)

MIXED-ORDER: $k' = k + 1$ ($= 2$)



● CELL DOFS (DEGREE k')

◆ FACE DOFS (DEGREE k)

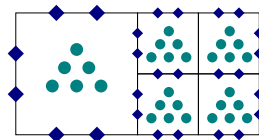
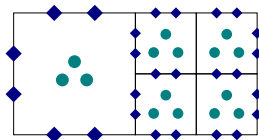
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- **Global degrees of freedom:** $\hat{u}_{\mathcal{M}} := (u_{\mathcal{T}}, u_{\mathcal{F}}) \in \hat{P}_{\mathcal{M}}^k := P_{\mathcal{T}}^{k'} \times P_{\mathcal{F}}^k$

▶ Cell unknowns: $u_{\mathcal{T}} \in P_{\mathcal{T}}^{k'} := \bigtimes_{T \in \mathcal{T}} P^{k'}(T)$

▶ Face unknowns: $u_{\mathcal{F}} \in P_{\mathcal{F}}^k := \bigtimes_{F \in \mathcal{F}} P^k(F)$

- Algebraic realization for elliptic problem: $(\kappa \nabla u, \nabla w)_\Omega = (f, w)_\Omega$

$$\begin{bmatrix} \mathcal{K}_\mathcal{T} & \mathcal{K}_{\mathcal{T}\mathcal{F}} \\ \mathcal{K}_{\mathcal{F}\mathcal{T}} & \mathcal{K}_\mathcal{F} \end{bmatrix} \begin{bmatrix} \mathbf{U}_\mathcal{T} \\ \mathbf{U}_\mathcal{F} \end{bmatrix} = \begin{bmatrix} \mathbf{F}_\mathcal{T} \\ 0 \end{bmatrix}$$

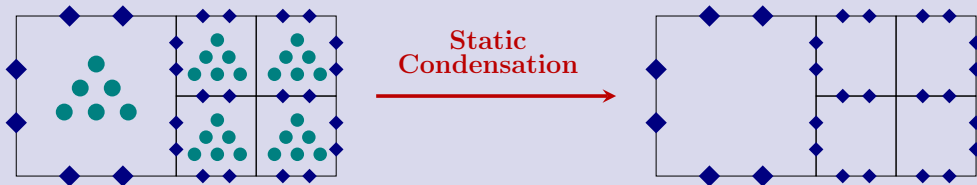
- Coupling between cell and face unknowns: $\mathcal{K}_\mathcal{F} \mathbf{U}_\mathcal{F} = -\mathcal{K}_{\mathcal{F}\mathcal{T}} \mathbf{U}_\mathcal{T}$

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- Static condensation: Elimination of cell unknowns



■ Design of local operators

- **Gradient reconstruction** $(\nabla u)|_T \approx \mathbf{g}_T(\hat{u}_T) \in \mathbf{P}^k(T)$: design mimics **integration by parts**

$$\forall \hat{u}_T \in \hat{P}_T^k \quad (\mathbf{g}_T(\hat{u}_T), \mathbf{q})_T = (\nabla u_T, \mathbf{q})_T - (u_T - u_{\partial T}, \mathbf{q} \cdot \mathbf{n}_T)_{\partial T}, \quad \forall \mathbf{q} \in \mathbf{P}^k(T)$$

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- **Stabilization operator**: Weakly enforces matching of trace of cell dofs with face dofs

$$S_{\partial T}(\hat{u}_T) := \begin{cases} \Pi_{\partial T}^k(u_T|_{\partial T} - u_{\partial T}) & \text{mixed-order } (k' = k + 1): \text{ Lehrenfeld-Schöberl (HDG)} \\ u_T|_{\partial T} - u_{\partial T} + \text{HOC} & \text{equal-order } (k' = k): \text{ Specific to HHO methods} \end{cases}$$

HIGH-ORDER CORRECTION TERMS



■ Same advantages as dG with following improvements:

- ✓ Tuning-free and cell-based stabilization
- ✓ No integration needed on faces for nonlinear problems

► Explicit discretizations:

- ✓ Memory cost comparable to dG:

Face dofs represent only a small fraction of the total dofs in the discretization

► Implicit discretizations:

- ✓ Improved convergence rates: $\|\cdot\|_{H^1}: \mathcal{O}(h^{k+1})$ vs. $\mathcal{O}(h^k)$ & $\|\cdot\|_{L^2}: \mathcal{O}(h^{k+2})$ vs. $\mathcal{O}(h^{k+1})$
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Hybrid nonconforming methods are a versatile tool allowing both efficient explicit and implicit discretizations!

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II.3 Runge–Kutta time discretization

II.4 Numerical results



R. Mottier, A. Ern, R. Khot, L. Guillot (2025). Accepted to M2AN

Hybrid high-order methods for elasto-acoustic wave propagation in the time domain



R. Mottier, A. Ern, L. Guillot (2025). Submitted to CMAME

Elasto-acoustic wave propagation in geophysical media using hybrid high-order methods on general meshes

II.1. Model problem

- Wave equations in first-order formulation in time → **Wide range of efficient time integrators**

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- **Acoustic wave equation (in Ω^F):**

$$\rho^F \partial_t \mathbf{v} - \nabla p = \mathbf{0}$$

$$\frac{1}{\kappa} \partial_t p - \nabla \cdot \mathbf{v} = f^F$$

Unknowns: p pressure, $\mathbf{v}$ velocity

Parameters: $\rho^F, \kappa \longrightarrow c_p^F := \sqrt{\frac{\kappa}{\rho^F}}$

- **Elastic wave equation (in Ω^S):**

$$\mathbb{C}^{-1}(\lambda, \mu) \partial_t \mathbb{s} - \nabla_{\text{sym}} \mathbf{v} = \mathbf{0}$$

$$\rho^S \partial_t \mathbf{v} - \nabla \cdot \mathbb{s} = \mathbf{f}^S$$

Unknowns: $\mathbf{v}$ velocity, $\mathbb{s}$ stress

Parameters: $\rho^S, \mathbb{C} \longrightarrow c_p^S := \sqrt{\frac{\lambda + 2\mu}{\rho^S}}, \quad c_s := \sqrt{\frac{\mu}{\rho^S}}$

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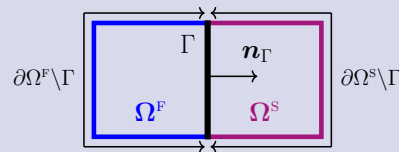
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- ▶ **Coupling conditions (on $\Gamma = \partial\Omega^F \cap \partial\Omega^S$):**

▶ Kinematic condition: $\mathbf{v} \cdot \mathbf{n}_\Gamma = \mathbf{v} \cdot \mathbf{n}_\Gamma$

▶ Balance of forces per unit surface: $\mathbb{s} \cdot \mathbf{n}_\Gamma = p \mathbf{n}_\Gamma$



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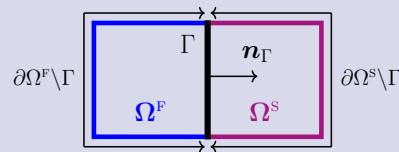
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- Initial conditions on $(p, \mathbf{v})$ and $(\mathbf{v}, \mathbb{s})$
- Homogeneous Dirichlet boundary conditions on $\partial\Omega$ for simplicity

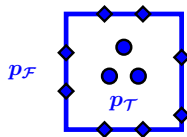
HHO

■ Main idea:

Primal variables p

■ Local dofs:

$$\hat{p}_T := (p_T, p_{\partial T}) \in P^{k'}(T) \times P^k(F)$$



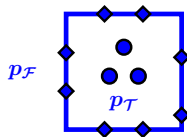
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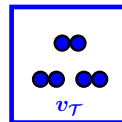
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dG

Dual variables v

$$v_T \in P^k(T)$$



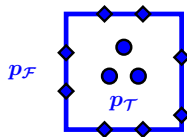
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■ Global dofs:

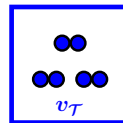
$$\hat{p}_{\mathcal{M}^F} := (p_{\mathcal{T}^F}, p_{\mathcal{F}^F}) \in \hat{P}_{\mathcal{M}^F}^k,$$

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dG

Dual variables v

$$v_T \in P^k(T)$$



$$v_{\mathcal{T}^F} \in P^k(\mathcal{T})$$

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II.2. HHO space semi-discretization

- **Acoustic weak formulation:** Find $(\mathbf{v}, p) : (0, T_f) \rightarrow \mathbf{M}^F \times P^F$ such that, $\forall t \in (0, T_f)$,

$$(\partial_t \mathbf{v}(t), \mathbf{r})_{(\rho^F; \Omega^F)} - (\nabla p(t), \mathbf{r})_{\Omega^F} = 0$$

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- **Acoustic wave equation:** Find $(\hat{p}_{\mathcal{M}^F}, \mathbf{v}_{\mathcal{T}^F}) : (0, T_f) \longrightarrow \hat{P}_{\mathcal{M}^F}^k \times \mathbf{V}_{\mathcal{T}^F}^k$ such that, $\forall t \in (0, T_f)$,

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- **Global operators:**

$$\triangleright \mathbf{g}_{\mathcal{T}^F}(\hat{p}_{\mathcal{M}^F}(t))|_T := \mathbf{g}_T(\hat{p}_T)$$

$$\triangleright s_{\mathcal{M}^F}(\hat{p}_{\mathcal{M}^F}, \hat{q}_{\mathcal{M}^F}) := \sum_{T \in \mathcal{T}^F} \tau_T^F(S_{\partial T}(\hat{p}_T), S_{\partial T}(\hat{q}_T))_{\partial T}$$

II.2. HHO space semi-discretization

- **Acoustic weak formulation:** Find $(\mathbf{v}, p) : (0, T_f) \rightarrow \mathbf{M}^F \times P^F$ such that, $\forall t \in (0, T_f)$,

$$(\partial_t \mathbf{v}(t), \mathbf{r})_{(\rho^F; \Omega^F)} - (\nabla p(t), \mathbf{r})_{\Omega^F} = 0$$

$$(\partial_t p(t), q)_{(\frac{1}{\kappa}; \Omega^F)} + (\mathbf{v}(t), \nabla q)_{\Omega^F} = (f^F(t), q)_{\Omega^F}$$

- **Acoustic wave equation:** Find $(\hat{p}_{\mathcal{M}^F}, \mathbf{v}_{\mathcal{T}^F}) : (0, T_f) \longrightarrow \hat{P}_{\mathcal{M}^F}^k \times \mathbf{V}_{\mathcal{T}^F}^k$ such that, $\forall t \in (0, T_f)$,

$$(\partial_t \mathbf{v}_{\mathcal{T}^F}(t), \mathbf{r}_{\mathcal{T}^F})_{(\rho^F; \Omega^F)} - (\mathbf{g}_{\mathcal{T}^F}(\hat{p}_{\mathcal{M}^F}(t)), \mathbf{r}_{\mathcal{T}^F})_{\Omega^F} = 0$$

$$(\partial_t p_{\mathcal{T}^F}(t), q_{\mathcal{T}^F})_{(\frac{1}{\kappa}; \Omega^F)} + (\mathbf{v}_{\mathcal{T}^F}(t), \mathbf{g}_{\mathcal{T}^F}(\hat{q}_{\mathcal{M}^F}))_{\Omega^F} + s_{\mathcal{M}^F}(\hat{p}_{\mathcal{M}^F}(t), \hat{q}_{\mathcal{M}^F}) = (f^F(t), q_{\mathcal{T}^F})_{\Omega^F}$$

- **Global operators:**

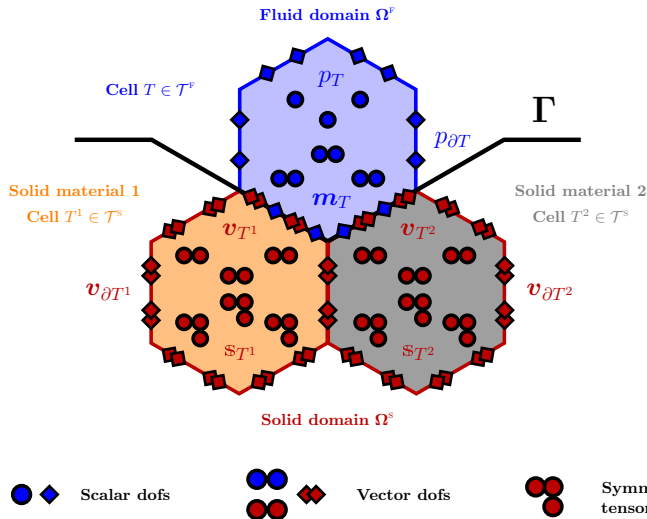
$$\triangleright \mathbf{g}_{\mathcal{T}^F}(\hat{p}_{\mathcal{M}^F}(t))|_T := \mathbf{g}_T(\hat{p}_T)$$

$$\triangleright s_{\mathcal{M}^F}(\hat{p}_{\mathcal{M}^F}, \hat{q}_{\mathcal{M}^F}) := \sum_{T \in \mathcal{T}^F} \tau_T^F(S_{\partial T}(\hat{p}_T), S_{\partial T}(\hat{q}_T))_{\partial T}$$

- **Error analysis**  Mottier, Ern, Khot, Guillot (2025)

$$\|\mathbf{v} - \mathbf{v}_{\mathcal{T}^F}\|_{\rho^F; \Omega^F} + \|p - p_{\mathcal{T}^F}\|_{\frac{1}{\kappa}; \Omega^F} \lesssim \begin{cases} \mathcal{O}(h^{k+\frac{1}{2}}) & \text{on general meshes} \\ \mathcal{O}(h^{k+1}) & \text{on simplices with HDG projections} \end{cases}$$

DOFs of the HHO space semi-discretization of the elasto-acoustic coupling problem



Runge–Kutta time integration

■ Compact formulation of the space semi-discrete algebraic realization

$$\begin{bmatrix} \mathcal{M}_{\mathcal{T}} & 0 \\ 0 & 0 \end{bmatrix} \frac{d}{dt} \begin{bmatrix} \mathbf{U}_{\mathcal{T}} \\ \mathbf{U}_{\mathcal{F}} \end{bmatrix} + \begin{bmatrix} \mathcal{K}_{\mathcal{T}} & \mathcal{K}_{\mathcal{T}\mathcal{F}} \\ \mathcal{K}_{\mathcal{F}\mathcal{T}} & \mathcal{K}_{\mathcal{F}} \end{bmatrix} \begin{bmatrix} \mathbf{U}_{\mathcal{T}} \\ \mathbf{U}_{\mathcal{F}} \end{bmatrix} = \begin{bmatrix} \mathbf{F}_{\mathcal{T}} \\ 0 \end{bmatrix}$$

- ▶ $\mathcal{K}_{\mathcal{T}}$ is trivially block-diagonal
- ▶ $\mathcal{K}_{\mathcal{F}}$ is block-diagonal under certain conditions

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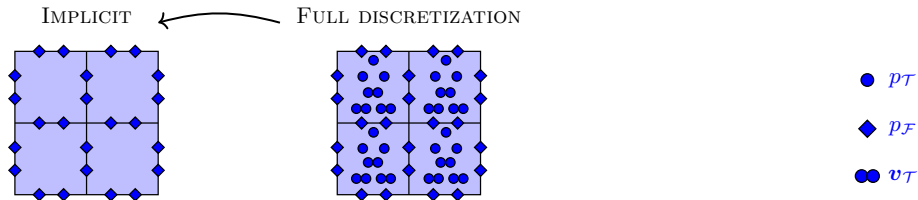
■ s -stage Runge–Kutta schemes

SDIRK($s, s+1$) : Truncation order: $\mathcal{O}(\Delta t^{s+1})$

c_1	a_{*}	0	$\cdots$	0
c_2	a_{21}	a_{*}	$\ddots$	0
$\vdots$	$\vdots$	$\ddots$	$\ddots$	$\vdots$
c_s	a_{s1}	$\cdots$	$a_{s,s-1}$	a_{*}
	b_1	$\cdots$	b_{s-1}	b_s

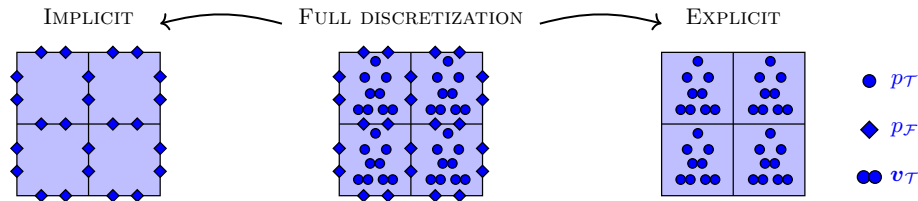
ERK(s) : Truncation order: $\mathcal{O}(\Delta t^s)$

c_1	0	$\cdots$	$\cdots$	0
c_2	a_{21}	0	$\cdots$	0
$\vdots$	$\vdots$	$\ddots$	$\ddots$	$\vdots$
c_s	a_{s1}	$\cdots$	$a_{s,s-1}$	0
	b_1	$\cdots$	b_{s-1}	b_s



■ Implicit discretization

- ▶ Elimination of the cell unknowns: **more than 80% of the system size**
- ▶ Superconvergence of HHO unknown $(k + 2)$ and classical rates for dG unknowns $(k + 1)$
- ▶ Competitive with respect to explicit methods in term of CPU time 📄 [Mottier, Ern, Guillot \(2026\)](#)



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■ Explicit discretization: dG rewritting

- ▶ Elimination of the face unknowns
- ▶ Classical convergence rates ($k + 1$) for HHO and dG unknowns
- ▶ **dG-like discretization**

■ **SDIRK($s, s + 1$) schemes:** For all $i \in \{1, \dots, s\}$,

$$\begin{bmatrix} \mathcal{M}_{\mathcal{T}} & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \mathbf{U}_{\mathcal{T}}^{n,i} \\ \mathbf{U}_{\mathcal{F}}^{n,i} \end{bmatrix} = \begin{bmatrix} \mathcal{M}_{\mathcal{T}} & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \mathbf{U}_{\mathcal{T}}^{n-1} \\ \mathbf{U}_{\mathcal{F}}^{n-1} \end{bmatrix} + \Delta t \sum_{j=1}^i a_{ij} \left(\begin{bmatrix} \mathbf{F}_{\mathcal{T}}^{n-1+c_j} \\ 0 \end{bmatrix} - \begin{bmatrix} \mathcal{K}_{\mathcal{T}} & \mathcal{K}_{\mathcal{T}\mathcal{F}} \\ \mathcal{K}_{\mathcal{F}\mathcal{T}} & \mathcal{K}_{\mathcal{F}} \end{bmatrix} \begin{bmatrix} \mathbf{U}_{\mathcal{T}}^{n,j} \\ \mathbf{U}_{\mathcal{F}}^{n,j} \end{bmatrix} \right)$$

► Each stage can be rewritten as

$$\begin{bmatrix} \mathcal{M}_{\mathcal{T}} + a_* \Delta t \mathcal{K}_{\mathcal{T}} & a_* \Delta t \mathcal{K}_{\mathcal{T}\mathcal{F}} \\ a_* \Delta t \mathcal{K}_{\mathcal{F}\mathcal{T}} & a_* \Delta t \mathcal{K}_{\mathcal{F}} \end{bmatrix} \begin{bmatrix} \mathbf{U}_{\mathcal{T}}^{n,i} \\ \mathbf{U}_{\mathcal{F}}^{n,i} \end{bmatrix} = \begin{bmatrix} \gamma_{\mathcal{T}}^{n,i} \\ \gamma_{\mathcal{F}}^{n,i} \end{bmatrix}$$

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■ **ERK(s) schemes:** For all $i \in \{1, \dots, s\}$,

$$\begin{bmatrix} \mathcal{M}_{\mathcal{T}} & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \mathbf{U}_{\mathcal{T}}^{n,i} \\ \mathbf{U}_{\mathcal{F}}^{n,i} \end{bmatrix} = \begin{bmatrix} \mathcal{M}_{\mathcal{T}} & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \mathbf{U}_{\mathcal{T}}^{n-1} \\ \mathbf{U}_{\mathcal{F}}^{n-1} \end{bmatrix} + \Delta t \sum_{j=1}^{\textcolor{red}{i}-1} a_{ij} \left(\begin{bmatrix} \mathbf{F}_{\mathcal{T}}^{n-1+c_j} \\ 0 \end{bmatrix} - \begin{bmatrix} \mathcal{K}_{\mathcal{T}} & \mathcal{K}_{\mathcal{T}\mathcal{F}} \\ \mathcal{K}_{\mathcal{F}\mathcal{T}} & \textcolor{red}{\mathcal{K}}_{\mathcal{F}} \end{bmatrix} \begin{bmatrix} \mathbf{U}_{\mathcal{T}}^{n,j} \\ \mathbf{U}_{\mathcal{F}}^{n,j} \end{bmatrix} \right)$$

► **Face dofs elimination:** $\textcolor{red}{\mathcal{K}}_{\mathcal{F}} \mathbf{U}_{\mathcal{F}}^{n,j} = -\mathcal{K}_{\mathcal{F}\mathcal{T}} \mathbf{U}_{\mathcal{T}}^{n,j}$

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$$\mathbf{U}_{\mathcal{T}}^{n,i} = \mathbf{U}_{\mathcal{T}}^{n-1} + \Delta t \sum_{j=1}^{i-1} a_{ij} \left(\mathcal{M}_{\mathcal{T}}^{-1} \mathbf{F}_{\mathcal{T}}^{n-1+c_j} + \mathcal{B}_{\mathcal{T}} \mathbf{U}_{\mathcal{T}}^{n,j} \right)$$

with $\mathcal{B}_{\mathcal{T}} := -\mathcal{M}_{\mathcal{T}}^{-1} (\mathcal{K}_{\mathcal{T}} - \mathcal{K}_{\mathcal{T}\mathcal{F}} \mathcal{K}_{\mathcal{F}}^{-1} \mathcal{K}_{\mathcal{F}\mathcal{T}})$ the **condensed system matrix**

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- Fully explicit scheme which handle general meshes with nonconformities
 $\implies$ mesh generation easier + less small elements
- Remaining small elements require strategies to handle the CFL restriction

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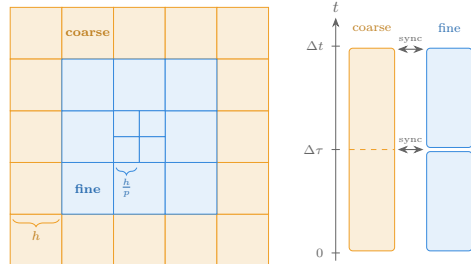
✓ Mesh generation easier & less small elements

✗ Remaining small elements require strategies to handle the CFL restriction

■ **LTS-ERK(s) schemes** 📄 Grote, Mehlin, Mitkova (2015)

- **Mesh partition:** Splitting DOFs into **coarse** (step Δt) and **fine** (step $\Delta\tau = \Delta t/p$) parts

$$U(t) = \underbrace{(\mathcal{I} - \mathcal{P}) U(t)}_{U^c(t)} + \underbrace{\mathcal{P} U(t)}_{U^F(t)}$$



- **ERK(s) coarse:** Approximation of the coarse evolution via Taylor expansion. For all $i \in \{1, \dots, s\}$

$$\begin{aligned}
 (i) \quad \mathbf{g}_{\mathcal{T}}^{(i)} &\leftarrow \mathcal{B}_{\mathcal{T}}^i \mathbf{U}_{\mathcal{T}}^n + \sum_{j=0}^{i-1} \mathcal{B}_{\mathcal{T}}^{i-1-j} \mathbf{F}_{\mathcal{T}}^j, & (ii) \quad \mathbf{w}_{\mathcal{T}}^{(i)} &\leftarrow -\mathcal{B}_{\mathcal{T}} (\mathcal{I} - \mathcal{P}) \mathbf{g}_{\mathcal{T}}^{(i)} + (\mathcal{I} - \mathcal{P}) \mathcal{M}_{\mathcal{T}}^{-1} \mathbf{F}_{\mathcal{T}}^i \\
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- **ERK(s) fine (on substeps):** Approximation of fine part + synchronization with coarse part.
For all $i \in \{1, \dots, s\}$,

$$\begin{aligned}
 (i) \quad \mathbf{y}^{(i)} &\leftarrow \tilde{\mathbf{U}}^{\frac{m}{p}} + c_i \mathbf{k}^{(i)} & (ii) \quad \mathbf{k}_{\mathcal{T}}^{(i)} &\leftarrow \mathcal{M}_{\mathcal{T}}^{-1} \left(\mathcal{P} \mathbf{F}_{\mathcal{T}}^{\tau_i} - \mathcal{K}_{\mathcal{T}} \mathcal{P} \mathbf{y}_{\mathcal{T}}^{(i)} - \mathcal{K}_{\mathcal{T}\mathcal{F}} \mathcal{P} \mathbf{y}_{\mathcal{F}}^{(i)} \right) \\
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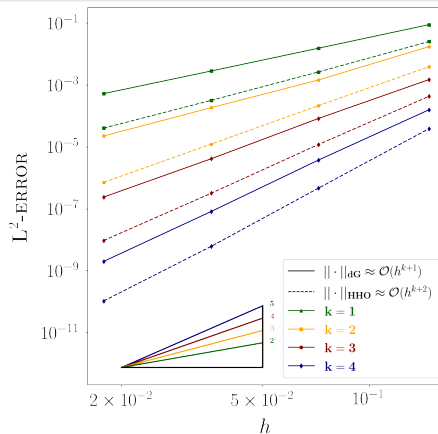
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- **Final update (on substeps):** $\tilde{\mathbf{U}}^{\frac{m+1}{p}} \leftarrow \tilde{\mathbf{U}}^{\frac{m}{p}} + \sum_{i=1}^s b_i \mathbf{k}^{(i)}$

Numerical results

■ **Convergence rates on general meshes:** $\|\cdot\|_{\text{dG}}: \mathcal{O}(h^{k+1})$ $\|\cdot\|_{\text{HHO}}: \mathcal{O}(h^{k+2})$

Implicit schemes

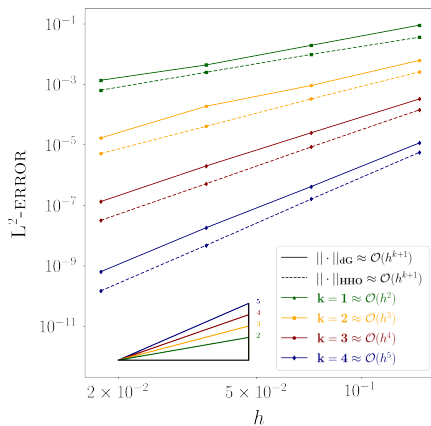


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ERK(s) schemes

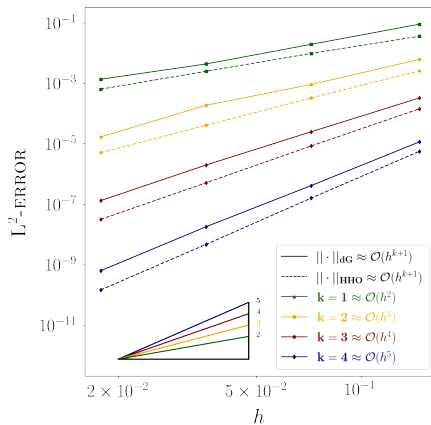


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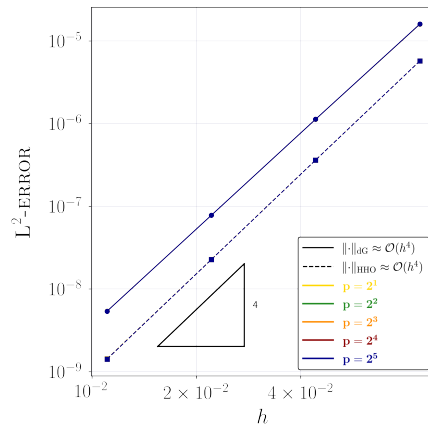
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ERK(s) schemes



LTS-ERK(4) scheme:

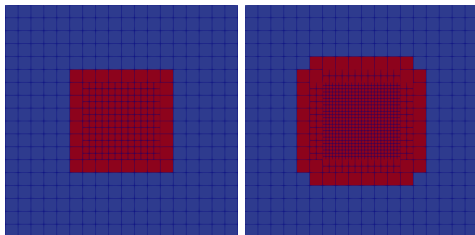


Stability of the LTS-ERK(4) scheme

- **Principle:** One LTS-RK4 time-step is **linear** and $\mathbf{U}_{\mathcal{T}}^{n+1} = \mathcal{C} \mathbf{U}_{\mathcal{T}}^n$ with $\mathcal{C}$ the **amplification matrix**
- **Construction:** Apply LTS-RK(4) to each vector $\mathbf{e}_i$ of the **canonical basis**
→ $\mathcal{C}$ built **column by column** (instead of using $\mathcal{C} = \mathcal{I} + \Delta t \mathcal{B}_{\mathcal{T}} + \frac{\Delta t^2}{2} \mathcal{B}_{\mathcal{T}}^2 + \frac{\Delta t^3}{3!} \mathcal{B}_{\mathcal{T}}^3 + \frac{\Delta t^4}{4!} \mathcal{B}_{\mathcal{T}}^4$)

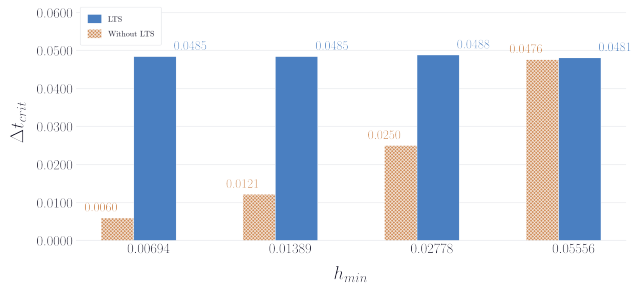
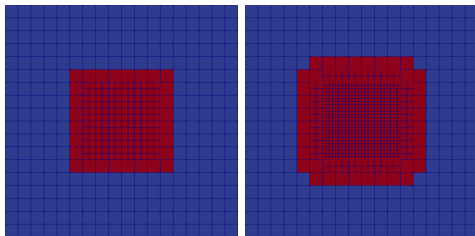
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- **Procedure:** For each **refinement level**, find the largest stable Δt (*i.e.* $\rho(\mathcal{C}) \leq 1$)
 for both **RK4 & LTS-RK4**



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Standard RK4: Δt halved at each refinement (*as expected*, $\Delta t = \mathcal{O}(h)$)

LTS-RK4: Δt remains constant, thanks to the local time-step $\Delta \tau$

Propagation of a Ricker wavelet in a bilayered domain

■ Test case setting

- ▶ $v_0(x, y) := \theta \exp\left(-\pi^2 \frac{r^2}{\lambda^2}\right) (x - x_c, y - y_c)^\dagger$
- ▶ Fluid subdomain: **Water**
- ▶ Solid subdomain: **Hardrock (Granite)**



Propagation of a Ricker wavelet in a bilayered domain

■ Test case setting

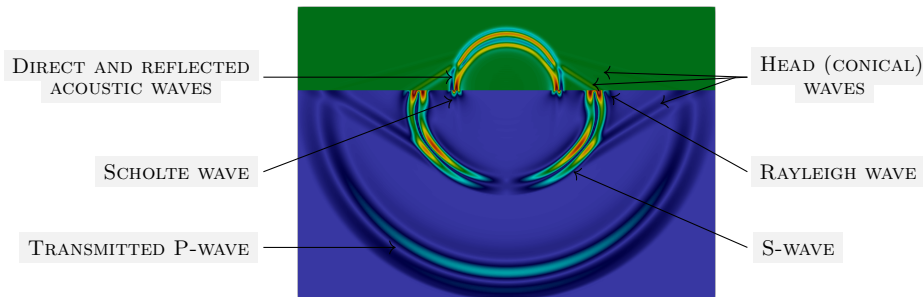
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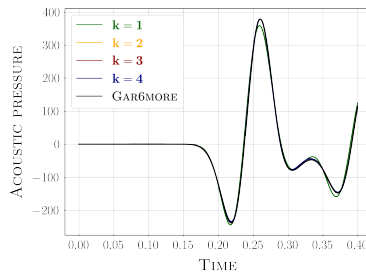
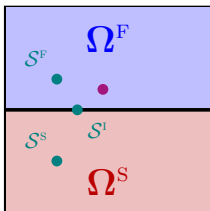
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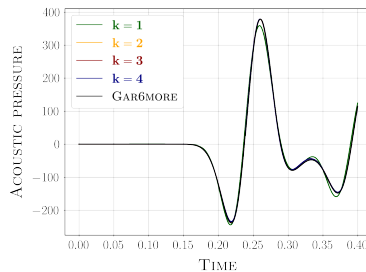
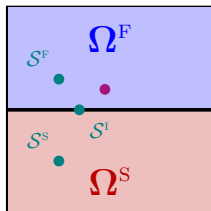
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■ Comparison to semi-analytical solution (Gar6more) Diaz, Ezziani (2008)



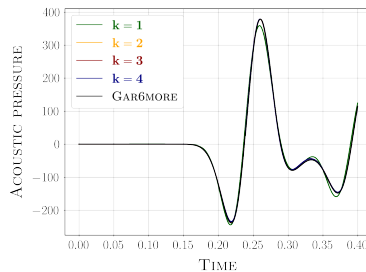
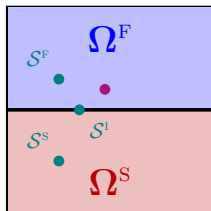
■ Comparison to semi-analytical solution (Gar6more) Diaz, Ezziani (2008)



■ Efficiency study: Fair comparison (for 3D perspective)

- ▶ Implicit schemes with **iterative** (BiCG) **solver** (with ILU preconditioner)
- ▶ Explicit schemes (without LTS) with **time step just below stability limit**

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- Explicit schemes (without LTS) with **time step just below stability limit**

► Implicit time steps:
almost $7\times$ larger

► Implicit schemes:
decrease by 25% CPU time

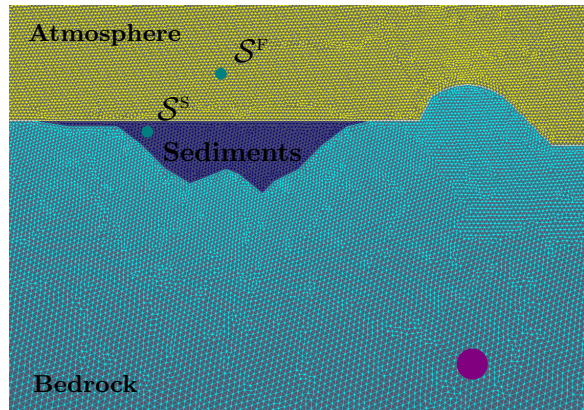
SCHEMES		SOLVER	CFL*	$c_{\#} \frac{\Delta t}{h}$	RATIO	CPU [s]	RATIO	ERR
SDIRK(3, 4)	$k = 2$	direct	n/a	0.414	6.90	1125	1	3.15e-02
	$k = 3$			0.414	6.90	2284	2.03	2.52e-02
SDIRK(3, 4)	$k = 2$	iterative	n/a	0.414	6.90	2228	1.98	3.91e-02
	$k = 3$			0.414	6.90	4216	3.75	3.11e-02
ERK(4)	$k = 2$	n/a	0.138	0.095	1.36	1533	1.27	2.01e-2
	$k = 3$		0.087	0.060	1	5664	5.03	1.90e-2

Propagation of a Ricker wavelet in a sedimentary basin

- Test case setting: typical geometry difficult to mesh in 3D

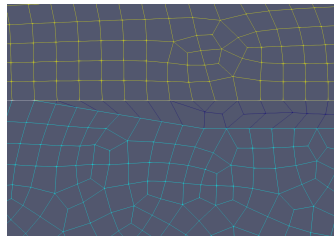
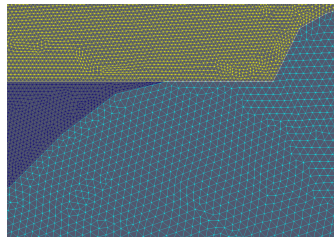
MATERIAL	$\rho^{F/S} \left[\frac{\text{kg}}{\text{m}^3} \right]$	$c_P^{F/S} \left[\frac{\text{m}}{\text{s}} \right]$	$c_P^S \left[\frac{\text{m}}{\text{s}} \right]$
ATMOSPHERE	1.225	343	n/a
SEDIMENTS	1300	1600	900
BEDROCK	2570	5350	3009

$$\mathbf{v}_0(x, y) := \theta \exp \left(-\pi^2 \frac{r^2}{\lambda^2} \right) (x - x_c, y - y_c)^\dagger$$



- **Elasto-acoustic coupling challenges:**
 - ▶ Difficulties in meshing the geometry of interest

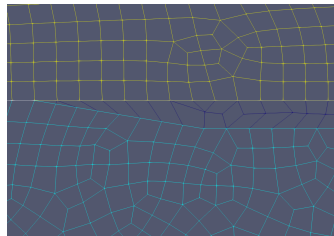
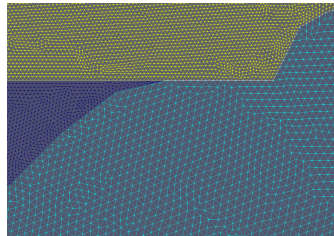
Mixed meshes

Nonconforming meshes
with hanging nodes

- **Elasto-acoustic coupling challenges:**
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- Interest of our approach:
 - ▶ Each subdomain can be meshed independently with different mesh shapes

Mixed meshes

Nonconforming meshes
with hanging nodes

■ Elasto-acoustic coupling challenges:

- ▶ Difficulties in meshing the geometry of interest
- ▶ **Large contrast in wave speeds:**

$$c_p^F \ll c_p^S \implies \lambda^F \ll \lambda^S$$

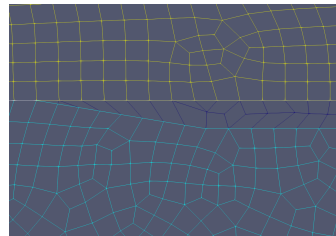
≈ 5 interpolation nodes per wavelenght

Acoustic subdomains need finner meshes

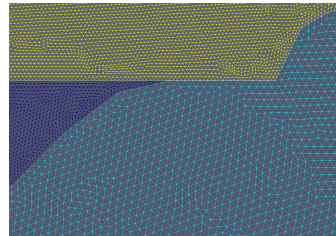
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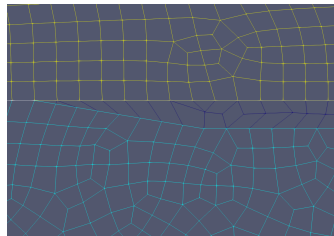
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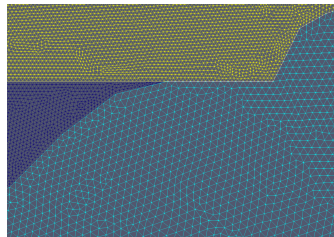
■ Interest of our approach:

- ▶ Each subdomain can be meshed independently with different mesh shapes
- ▶ Local time stepping used in the acoustic subdomain & i the fine cells of the solid region

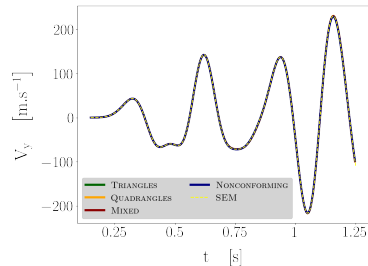
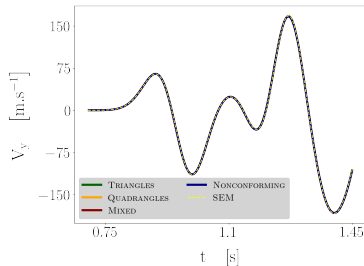
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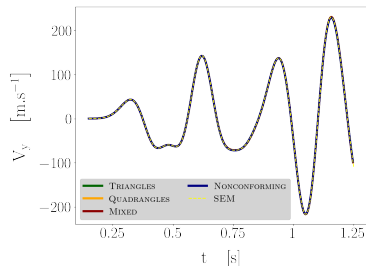
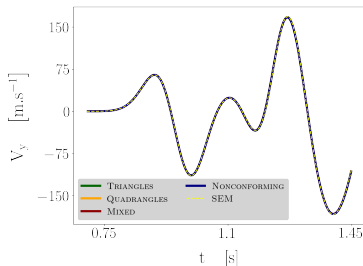
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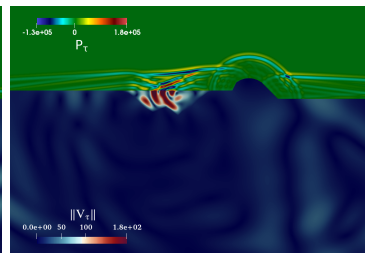
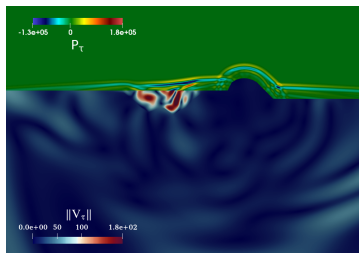
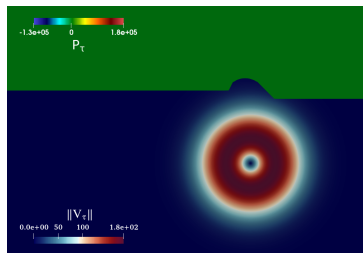
■ Comparison with a reference spectral solver



■ Comparison with a reference spectral solver



■ Wave trapped into sedimentary basin



- **Conclusion:** We developed a nonconforming hybrid method (HHO) which,
 - ▶ handle **general meshes**
 - ▶ have **efficient explicit** and **implicit** discretizations
 - ▶ have a **LTS** strategy for fully explicit discretizations.

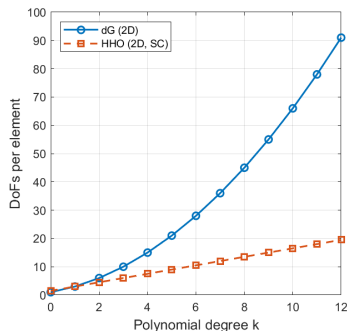
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- **Future work:**
 - ▶ Locally implicit time integration
 - ▶ Adaptive mesh refinement strategies combined with LTS-ERK schemes

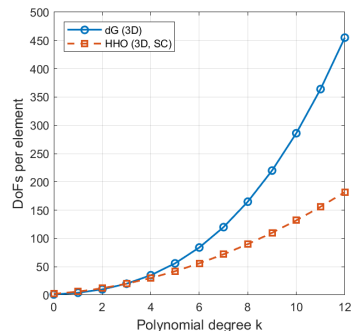
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THANK YOU FOR YOUR
ATTENTION!



2D; Triangle



3D; Tetrahedron

$$\frac{1}{2}(k+1)(k+2) \quad \text{dG}$$

$$\frac{3}{2}(k+1) \quad \text{HHO}$$

$$\frac{1}{6}(k+1)(k+2)(k+3) \quad \text{dG}$$

$$(k+1)(k+2) \quad \text{HHO}$$

After static condensation, HHO has dramatically fewer DoFs than dG for high-order approximations!

■ Approximation spaces

$$\begin{aligned} P^{\mathsf{F}} &:= \{p \in H^1(\Omega^{\mathsf{F}}) : p|_{\partial\Omega^{\mathsf{F}}\setminus\Gamma} = 0\}, & \boldsymbol{M}^{\mathsf{F}} &:= \boldsymbol{L}^2(\Omega^{\mathsf{F}}) \\ \boldsymbol{V}^{\mathsf{S}} &:= \{\boldsymbol{v} \in \boldsymbol{H}^1(\Omega^{\mathsf{S}}) : \boldsymbol{v}|_{\partial\Omega^{\mathsf{S}}\setminus\Gamma} = \mathbf{0}\}, & \mathbb{S}^{\mathsf{S}} &:= \mathbb{L}_{\text{sym}}^2(\Omega^{\mathsf{S}}) \end{aligned}$$

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■ **Weak formulation:** Find $(v, p) : (0, T_f) \rightarrow M^F \times P^F$ and $(s, v) : (0, T_f) \rightarrow \mathbb{S}^S \times V^S$ such that, $\forall t \in (0, T_f)$,

► **Acoustic wave equations:** $\forall (r, q) \in M^F \times P^F$,

$$\begin{aligned} (\partial_t v(t), r)_{\rho^F; \Omega^F} - (\nabla p(t), r)_{\Omega^F} &= 0 \\ (\partial_t p(t), q)_{\frac{1}{\kappa}; \Omega^F} + (v(t), \nabla q)_{\Omega^F} + (v(t) \cdot n_\Gamma, q)_\Gamma &= (f^F(t), q)_{\Omega^F} \end{aligned}$$

► **Elastic wave equations:** $\forall (l, w) \in \mathbb{S}^S \times V^S$,

$$\begin{aligned} (\partial_t s(t), l)_{\mathbb{C}^{-1}; \Omega^S} - (\nabla_{\text{sym}} v(t), l)_{\Omega^S} &= 0, \\ (\partial_t v(t), w)_{\rho^S; \Omega^S} + (s(t), \nabla_{\text{sym}} w)_{\Omega^S} - (p(t) n_\Gamma, w)_\Gamma &= (f^S(t), w)_{\Omega^S}. \end{aligned}$$

► Skew-symmetry of differential operators

► Coupling conditions weakly imposed

II.4. Numerical results

■ Implicit discretization

- ▶ $k' = k + 1$ (mixed-order discretization)
 - ▶ More unknowns but removed by static condensation
 - ▶ More efficient than equal-order with specific HHO stabilization
- ▶ Expected results: **superconvergence of the primal variable**

II.4. Numerical results

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■ Explicit discretization

- ▶ $k' = k$ (equal-order discretization) → Less degrees of freedom
- ▶ Nonstandard stabilization:
 1. Plain Least-Squares → Ensure block-diagonal structure
 2. $\mathcal{O}(1)$ -stabilization → Classical CFL condition: $\Delta t = \mathcal{O}(h)$
- ▶ Expected results: **classical convergence rates and CFL condition**

■ Influence of cell geometry

		$s = 2$			$s = 3$			$s = 4$		
MESHERS		$\triangle$	$\square$	$\hexagon$	$\triangle$	$\square$	$\hexagon$	$\triangle$	$\square$	$\hexagon$
$k = 1$	CFL*	0.191	0.205	0.264	0.238	0.253	0.329	0.265	0.282	0.363
	RATIO	1	1.07	1.38	1	1.06	1.38	1	1.06	1.37
$k = 2$	CFL*	0.106	0.099	0.136	0.133	0.123	0.170	0.147	0.138	0.188
	RATIO	1	0.93	1.28	1	0.92	1.28	1	0.94	1.28
$k = 3$	CFL*	0.072	0.063	0.082	0.090	0.079	0.102	0.100	0.087	0.115
	RATIO	1	0.88	1.13	1	0.88	1.13	1	0.87	1.15

- ▶ CFL* mildly sensitive to mesh geometry
- ▶ CFL improvement for meshes with higher face counts if $k = 1$
- ▶ Less pronounced improvement for $k \geq 2$

■ Space semi-discrete algebraic realization

$$\begin{bmatrix} \mathcal{M}_{\mathcal{T}^F}^{\rho^F} & 0 & 0 & 0 & 0 & 0 \\ 0 & \mathcal{M}_{\mathcal{T}^F}^{\frac{1}{\kappa}} & 0 & 0 & 0 & 0 \\ 0 & 0 & \mathcal{M}_{\mathcal{T}^S}^{\mathbb{C}^{-1}} & 0 & 0 & 0 \\ 0 & 0 & 0 & \mathcal{M}_{\mathcal{T}^S}^{\rho^S} & 0 & 0 \\ \hline 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \frac{d}{dt} \begin{bmatrix} \mathbf{M}_{\mathcal{T}^F} \\ \mathbf{P}_{\mathcal{T}^F} \\ \mathbf{S}_{\mathcal{T}^S} \\ \mathbf{V}_{\mathcal{T}^S} \\ \hline \mathbf{P}_{\mathcal{F}^F} \\ \mathbf{V}_{\mathcal{F}^S} \end{bmatrix} + \begin{bmatrix} 0 & \mathcal{G}_{\mathcal{T}^F} & 0 & 0 & \mathcal{G}_{\mathcal{T}^F \mathcal{F}^F} & 0 \\ -\mathcal{G}_{\mathcal{T}^F}^\dagger & \Sigma_{\mathcal{T}^F} & 0 & 0 & \Sigma_{\mathcal{T}^F \mathcal{F}^F} & 0 \\ 0 & 0 & 0 & \mathcal{H}_{\mathcal{T}^S} & 0 & \mathcal{H}_{\mathcal{T}^S \mathcal{F}^S} \\ 0 & 0 & -\mathcal{H}_{\mathcal{T}^S}^\dagger & \Sigma_{\mathcal{T}^S} & 0 & \Sigma_{\mathcal{T}^S \mathcal{F}^S} \\ \hline -\mathcal{G}_{\mathcal{T}^F \mathcal{F}^F}^\dagger & \Sigma_{\mathcal{T}^F \mathcal{F}^F}^\dagger & 0 & 0 & \Sigma_{\mathcal{F}^F} & \mathcal{C}_{\mathcal{F}^\Gamma} \\ 0 & 0 & -\mathcal{H}_{\mathcal{T}^S \mathcal{F}^S}^\dagger & \Sigma_{\mathcal{T}^S \mathcal{F}^S}^\dagger & -\mathcal{C}_{\mathcal{F}^\Gamma}^\dagger & \Sigma_{\mathcal{F}^S} \end{bmatrix} \begin{bmatrix} \mathbf{M}_{\mathcal{T}^F} \\ \mathbf{P}_{\mathcal{T}^F} \\ \mathbf{S}_{\mathcal{T}^S} \\ \mathbf{V}_{\mathcal{T}^S} \\ \hline \mathbf{P}_{\mathcal{F}^F} \\ \mathbf{V}_{\mathcal{F}^S} \end{bmatrix} = \begin{bmatrix} 0 \\ \mathbf{F}_{\mathcal{T}^F} \\ \hline 0 \\ \mathbf{F}_{\mathcal{T}^S} \\ \hline 0 \\ 0 \end{bmatrix}$$

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$$\begin{bmatrix} \mathcal{M}_{\mathcal{T}^F}^{\rho^F} & 0 & 0 & 0 & 0 & 0 \\ 0 & \mathcal{M}_{\mathcal{T}^F}^{\frac{1}{\kappa}} & 0 & 0 & 0 & 0 \\ 0 & 0 & \mathcal{M}_{\mathcal{T}^S}^{\mathbb{C}^{-1}} & 0 & 0 & 0 \\ 0 & 0 & 0 & \mathcal{M}_{\mathcal{T}^S}^{\rho^S} & 0 & 0 \\ \hline 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \frac{d}{dt} \begin{bmatrix} \mathbf{M}_{\mathcal{T}^F} \\ \mathbf{P}_{\mathcal{T}^F} \\ \mathbf{S}_{\mathcal{T}^S} \\ \mathbf{V}_{\mathcal{T}^S} \\ \hline \mathbf{P}_{\mathcal{F}^F} \\ \mathbf{V}_{\mathcal{F}^S} \end{bmatrix} + \begin{bmatrix} 0 & \mathcal{G}_{\mathcal{T}^F} & 0 & 0 & \mathcal{G}_{\mathcal{T}^F \mathcal{F}^F} & 0 \\ -\mathcal{G}_{\mathcal{T}^F}^\dagger & \Sigma_{\mathcal{T}^F} & 0 & 0 & \Sigma_{\mathcal{T}^F \mathcal{F}^F} & 0 \\ 0 & 0 & 0 & \mathcal{H}_{\mathcal{T}^S} & 0 & \mathcal{H}_{\mathcal{T}^S \mathcal{F}^S} \\ 0 & 0 & -\mathcal{H}_{\mathcal{T}^S}^\dagger & \Sigma_{\mathcal{T}^S} & 0 & \Sigma_{\mathcal{T}^S \mathcal{F}^S} \\ \hline -\mathcal{G}_{\mathcal{T}^F \mathcal{F}^F}^\dagger & \Sigma_{\mathcal{T}^F \mathcal{F}^F}^\dagger & 0 & 0 & \Sigma_{\mathcal{F}^F} & \mathcal{C}_{\mathcal{F}^\Gamma} \\ 0 & 0 & -\mathcal{H}_{\mathcal{T}^S \mathcal{F}^S}^\dagger & \Sigma_{\mathcal{T}^S \mathcal{F}^S}^\dagger & -\mathcal{C}_{\mathcal{F}^\Gamma}^\dagger & \Sigma_{\mathcal{F}^S} \end{bmatrix} \begin{bmatrix} \mathbf{M}_{\mathcal{T}^F} \\ \mathbf{P}_{\mathcal{T}^F} \\ \mathbf{S}_{\mathcal{T}^S} \\ \mathbf{V}_{\mathcal{T}^S} \\ \hline \mathbf{P}_{\mathcal{F}^F} \\ \mathbf{V}_{\mathcal{F}^S} \end{bmatrix} = \begin{bmatrix} 0 \\ \mathbf{F}_{\mathcal{T}^F} \\ \hline 0 \\ \mathbf{F}_{\mathcal{T}^S} \\ \hline 0 \\ 0 \end{bmatrix}$$

■ Compact formulation:

$$\begin{bmatrix} \mathcal{M}_{\mathcal{T}} & 0 \\ 0 & 0 \end{bmatrix} \frac{d}{dt} \begin{bmatrix} \mathbf{U}_{\mathcal{T}} \\ \mathbf{U}_{\mathcal{F}} \end{bmatrix} + \begin{bmatrix} \mathcal{K}_{\mathcal{T}} & \mathcal{K}_{\mathcal{T}\mathcal{F}} \\ \mathcal{K}_{\mathcal{F}\mathcal{T}} & \mathcal{K}_{\mathcal{F}} \end{bmatrix} \begin{bmatrix} \mathbf{U}_{\mathcal{T}} \\ \mathbf{U}_{\mathcal{F}} \end{bmatrix} = \begin{bmatrix} \mathbf{F}_{\mathcal{T}} \\ 0 \end{bmatrix}$$

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► $\mathcal{K}_{\mathcal{T}}$ is trivially block-diagonal

- Local blocks of size $\dim(\mathbf{P}^{k'}(\mathcal{T})) + \dim(\mathbf{P}^k(\mathcal{T}))$ or $\dim(\mathbf{P}^{k'}(\mathcal{T})) + \dim(\mathbb{P}^k(\mathcal{T}))$

► **Key question:** Is $\mathcal{K}_{\mathcal{F}}$ block-diagonal ? (**Important for explicit schemes**)

■ Structure of $\mathcal{K}_{\mathcal{F}}$:

$$\mathcal{K}_{\mathcal{F}} = \begin{bmatrix} \Sigma_{\mathcal{F}^F} & \mathcal{C}_{\mathcal{F}^\Gamma} \\ -\mathcal{C}_{\mathcal{F}^\Gamma} & \Sigma_{\mathcal{F}^S} \end{bmatrix} = \begin{bmatrix} \Sigma_{\mathcal{F}^{OF}} & 0 & 0 & 0 \\ 0 & \Sigma_{\mathcal{F}^{OS}} & 0 & 0 \\ 0 & 0 & \Sigma_{\mathcal{F}^\Gamma} & \mathcal{C}_{\mathcal{F}^\Gamma} \\ 0 & 0 & -\mathcal{C}_{\mathcal{F}^\Gamma} & \Sigma_{\mathcal{F}^\Gamma} \end{bmatrix}$$

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$$\mathcal{K}_{\mathcal{F}} = \begin{bmatrix} \Sigma_{\mathcal{F}^F} & \mathcal{C}_{\mathcal{F}^\Gamma} \\ -\mathcal{C}_{\mathcal{F}^\Gamma} & \Sigma_{\mathcal{F}^S} \end{bmatrix} = \begin{bmatrix} \Sigma_{\mathcal{F}^{OF}} & 0 & 0 & 0 \\ 0 & \Sigma_{\mathcal{F}^{OS}} & 0 & 0 \\ 0 & 0 & \Sigma_{\mathcal{F}^\Gamma} & \mathcal{C}_{\mathcal{F}^\Gamma} \\ 0 & 0 & -\mathcal{C}_{\mathcal{F}^\Gamma} & \Sigma_{\mathcal{F}^\Gamma} \end{bmatrix}$$

- $\Sigma_{\mathcal{F}^F}$ and $\Sigma_{\mathcal{F}^S}$ block-diagonal only for **Lehrenfeld–Schöberl** and **Least-squares** stabilizations

$$S_{\partial T}(\hat{p}_T) := \begin{cases} \Pi_{\partial T}^k(p_T|_{\partial T} - p_{\partial T}) & \text{mixed-order: Lehrenfeld–Schöberl (HDG)} \\ p_T|_{\partial T} - p_{\partial T} + H_{\partial T} C & \text{equal-order: Plain Least-squares} \end{cases}$$

■ Structure of $\mathcal{K}_{\mathcal{F}}$:

$$\mathcal{K}_{\mathcal{F}} = \begin{bmatrix} \Sigma_{\mathcal{F}^F} & \mathcal{C}_{\mathcal{F}^\Gamma} \\ -\mathcal{C}_{\mathcal{F}^\Gamma} & \Sigma_{\mathcal{F}^S} \end{bmatrix} = \begin{bmatrix} \Sigma_{\mathcal{F}^{OF}} & 0 & 0 & 0 \\ 0 & \Sigma_{\mathcal{F}^{OS}} & 0 & 0 \\ 0 & 0 & \Sigma_{\mathcal{F}^\Gamma} & \mathcal{C}_{\mathcal{F}^\Gamma} \\ 0 & 0 & -\mathcal{C}_{\mathcal{F}^\Gamma} & \Sigma_{\mathcal{F}^\Gamma} \end{bmatrix}$$

$$\begin{bmatrix} \Sigma_{F_1^\Gamma} & \mathcal{C}_{F_1^\Gamma} & 0 & 0 & 0 \\ -\mathcal{C}_{F_1^\Gamma} & \Sigma_{F_1^\Gamma} & 0 & \vdots & \vdots \\ 0 & 0 & \ddots & 0 & 0 \\ \vdots & \vdots & 0 & \Sigma_{F_n^\Gamma} & \mathcal{C}_{F_n^\Gamma} \\ 0 & 0 & 0 & -\mathcal{C}_{F_n^\Gamma} & \Sigma_{F_n^\Gamma} \end{bmatrix}$$

- $\Sigma_{\mathcal{F}^F}$ and $\Sigma_{\mathcal{F}^S}$ block-diagonal only for **Lehrenfeld–Schöberl** and **Least-squares** stabilizations

$$S_{\partial T}(\hat{p}_T) := \begin{cases} \Pi_{\partial T}^k(p_T|_{\partial T} - p_{\partial T}) & \text{mixed-order: Lehrenfeld–Schöberl (HDG)} \\ p_T|_{\partial T} - p_{\partial T} + \mathbf{H} \times \mathbf{C} & \text{equal-order: Plain Least-squares} \end{cases}$$

- **Coupling terms at interface:** grouping acoustic and elastic dofs of each face

→ $\mathcal{K}_{\mathcal{F}}$ is block diagonal

■ **SDIRK($s, s + 1$) schemes:** For all $i \in \{1, \dots, s\}$,

$$\begin{bmatrix} \mathcal{M}_{\mathcal{T}} & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \mathbf{U}_{\mathcal{T}}^{n,i} \\ \mathbf{U}_{\mathcal{F}}^{n,i} \end{bmatrix} = \begin{bmatrix} \mathcal{M}_{\mathcal{T}} & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \mathbf{U}_{\mathcal{T}}^{n-1} \\ \mathbf{U}_{\mathcal{F}}^{n-1} \end{bmatrix} + \Delta t \sum_{j=1}^i a_{ij} \left(\begin{bmatrix} \mathbf{F}_{\mathcal{T}}^{n-1+c_j} \\ 0 \end{bmatrix} - \begin{bmatrix} \mathcal{K}_{\mathcal{T}} & \mathcal{K}_{\mathcal{T}\mathcal{F}} \\ \mathcal{K}_{\mathcal{F}\mathcal{T}} & \mathcal{K}_{\mathcal{F}} \end{bmatrix} \begin{bmatrix} \mathbf{U}_{\mathcal{T}}^{n,j} \\ \mathbf{U}_{\mathcal{F}}^{n,j} \end{bmatrix} \right)$$

► Each stage can be rewritten as

$$\begin{bmatrix} \mathcal{M}_{\mathcal{T}} + a_* \Delta t \mathcal{K}_{\mathcal{T}} & a_* \Delta t \mathcal{K}_{\mathcal{T}\mathcal{F}} \\ a_* \Delta t \mathcal{K}_{\mathcal{F}\mathcal{T}} & a_* \Delta t \mathcal{K}_{\mathcal{F}} \end{bmatrix} \begin{bmatrix} \mathbf{U}_{\mathcal{T}}^{n,i} \\ \mathbf{U}_{\mathcal{F}}^{n,i} \end{bmatrix} = \begin{bmatrix} \mathbf{B}_{\mathcal{T}}^{n,i} \\ \mathbf{B}_{\mathcal{F}}^{n,i} \end{bmatrix}$$

■ **SDIRK($s, s + 1$) schemes:** For all $i \in \{1, \dots, s\}$,

$$\begin{bmatrix} \mathcal{M}_{\mathcal{T}} & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \mathbf{U}_{\mathcal{T}}^{n,i} \\ \mathbf{U}_{\mathcal{F}}^{n,i} \end{bmatrix} = \begin{bmatrix} \mathcal{M}_{\mathcal{T}} & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \mathbf{U}_{\mathcal{T}}^{n-1} \\ \mathbf{U}_{\mathcal{F}}^{n-1} \end{bmatrix} + \Delta t \sum_{j=1}^i a_{ij} \left(\begin{bmatrix} \mathbf{F}_{\mathcal{T}}^{n-1+c_j} \\ 0 \end{bmatrix} - \begin{bmatrix} \mathcal{K}_{\mathcal{T}} & \mathcal{K}_{\mathcal{T}\mathcal{F}} \\ \mathcal{K}_{\mathcal{F}\mathcal{T}} & \mathcal{K}_{\mathcal{F}} \end{bmatrix} \begin{bmatrix} \mathbf{U}_{\mathcal{T}}^{n,j} \\ \mathbf{U}_{\mathcal{F}}^{n,j} \end{bmatrix} \right)$$

► Each stage can be rewritten as

$$\begin{bmatrix} \mathcal{M}_{\mathcal{T}} + a_* \Delta t \mathcal{K}_{\mathcal{T}} & a_* \Delta t \mathcal{K}_{\mathcal{T}\mathcal{F}} \\ a_* \Delta t \mathcal{K}_{\mathcal{F}\mathcal{T}} & a_* \Delta t \mathcal{K}_{\mathcal{F}} \end{bmatrix} \begin{bmatrix} \mathbf{U}_{\mathcal{T}}^{n,i} \\ \mathbf{U}_{\mathcal{F}}^{n,i} \end{bmatrix} = \begin{bmatrix} \mathbf{B}_{\mathcal{T}}^{n,i} \\ \mathbf{B}_{\mathcal{F}}^{n,i} \end{bmatrix}$$

■ **Usual static condensation: Cell dofs elimination**

- Global system **coupling only face dofs**
- **Cheap** local postprocessing to recover cell dofs

■ **ERK(s) schemes:** For all $i \in \{1, \dots, s\}$,

$$\begin{bmatrix} \mathcal{M}_{\mathcal{T}} & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \mathbf{U}_{\mathcal{T}}^{n,i} \\ \mathbf{U}_{\mathcal{F}}^{n,i} \end{bmatrix} = \begin{bmatrix} \mathcal{M}_{\mathcal{T}} & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \mathbf{U}_{\mathcal{T}}^{n-1} \\ \mathbf{U}_{\mathcal{F}}^{n-1} \end{bmatrix} + \Delta t \sum_{j=1}^{i-1} a_{ij} \left(\begin{bmatrix} \mathbf{F}_{\mathcal{T}}^{n-1+c_j} \\ 0 \end{bmatrix} - \begin{bmatrix} \mathcal{K}_{\mathcal{T}} & \mathcal{K}_{\mathcal{T}\mathcal{F}} \\ \mathcal{K}_{\mathcal{F}\mathcal{T}} & \mathcal{K}_{\mathcal{F}} \end{bmatrix} \begin{bmatrix} \mathbf{U}_{\mathcal{T}}^{n,j} \\ \mathbf{U}_{\mathcal{F}}^{n,j} \end{bmatrix} \right)$$

■ **Static condensation:**

► **Face dofs elimination:** $\mathcal{K}_{\mathcal{F}} \mathbf{U}_{\mathcal{F}}^{n,j} = -\mathcal{K}_{\mathcal{F}\mathcal{T}} \mathbf{U}_{\mathcal{T}}^{n,j}$

■ **ERK(s) schemes:** For all $i \in \{1, \dots, s\}$,

$$\begin{bmatrix} \mathcal{M}_{\mathcal{T}} & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} U_{\mathcal{T}}^{n,i} \\ U_{\mathcal{F}}^{n,i} \end{bmatrix} = \begin{bmatrix} \mathcal{M}_{\mathcal{T}} & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} U_{\mathcal{T}}^{n-1} \\ U_{\mathcal{F}}^{n-1} \end{bmatrix} + \Delta t \sum_{j=1}^{i-1} a_{ij} \left(\begin{bmatrix} F_{\mathcal{T}}^{n-1+c_j} \\ 0 \end{bmatrix} - \begin{bmatrix} \mathcal{K}_{\mathcal{T}} & \mathcal{K}_{\mathcal{T}\mathcal{F}} \\ \mathcal{K}_{\mathcal{F}\mathcal{T}} & \mathcal{K}_{\mathcal{F}} \end{bmatrix} \begin{bmatrix} U_{\mathcal{T}}^{n,j} \\ U_{\mathcal{F}}^{n,j} \end{bmatrix} \right)$$

■ **Static condensation:**

► **Face dofs elimination:** $\mathcal{K}_{\mathcal{F}} U_{\mathcal{F}}^{n,j} = -\mathcal{K}_{\mathcal{F}\mathcal{T}} U_{\mathcal{T}}^{n,j}$

► Each stage becomes (**dG rewriting by static condensation**):

$$\mathcal{M}_{\mathcal{T}} U_{\mathcal{T}}^{n,i} = \mathcal{M}_{\mathcal{T}} U_{\mathcal{T}}^{n-1} + \Delta t \sum_{j=1}^{i-1} a_{ij} \left(F_{\mathcal{T}}^{n-1+c_j} - \left(\mathcal{K}_{\mathcal{T}} - \mathcal{K}_{\mathcal{T}\mathcal{F}} \mathcal{K}_{\mathcal{F}}^{-1} \mathcal{K}_{\mathcal{F}\mathcal{T}} \right) U_{\mathcal{T}}^{n,j} \right)$$

■ Impact of stabilization on CFL stability limit:

$$c_{\sharp} \frac{\Delta t}{h} \leq \mathbf{CFL}^*(s, k, \eta^{\mathbf{f}}, \eta^{\mathbf{s}})$$

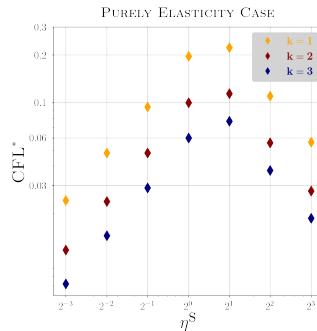
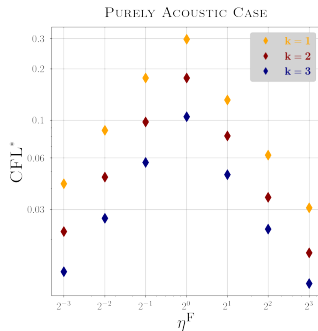
■ Impact of stabilization on CFL stability limit:

$$c_{\sharp} \frac{\Delta t}{h} \leq \mathbf{CFL}^*(s, k, \eta^f, \eta^s)$$

- Weighted $\mathcal{O}(1)$ -stabilization with $\eta^{f/s} := 2^m$ with $m \in \llbracket -3, 3 \rrbracket$

$$\tilde{s}_{\mathcal{M}^F}(\hat{p}_{\mathcal{M}^F}, \hat{q}_{\mathcal{M}^F}) := \eta^F s_{\mathcal{M}^F}(\hat{p}_{\mathcal{M}^F}, \hat{q}_{\mathcal{M}^F}),$$

$$\tilde{s}_{\mathcal{M}^S}(\hat{v}_{\mathcal{M}^S}, \hat{w}_{\mathcal{M}^S}) := \eta^S s_{\mathcal{M}^S}(\hat{v}_{\mathcal{M}^S}, \hat{w}_{\mathcal{M}^S})$$

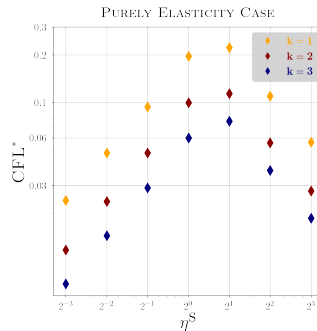
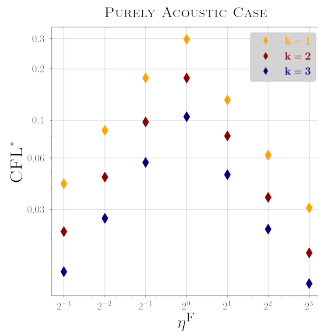


■ Impact of stabilization on CFL stability limit:

$$c_{\sharp} \frac{\Delta t}{h} \leq \mathbf{CFL}^*(s, k, \eta^f, \eta^s)$$

- Weighted $\mathcal{O}(1)$ -stabilization with $\eta^{f/s} := 2^m$ with $m \in \llbracket -3, 3 \rrbracket$

$$\tilde{s}_{\mathcal{M}^F}(\hat{p}_{\mathcal{M}^F}, \hat{q}_{\mathcal{M}^F}) := \eta^F s_{\mathcal{M}^F}(\hat{p}_{\mathcal{M}^F}, \hat{q}_{\mathcal{M}^F}), \quad \tilde{s}_{\mathcal{M}^S}(\hat{v}_{\mathcal{M}^S}, \hat{w}_{\mathcal{M}^S}) := \eta^S s_{\mathcal{M}^S}(\hat{v}_{\mathcal{M}^S}, \hat{w}_{\mathcal{M}^S})$$



- Evolution of $\mathbf{CFL}^*$ as $\min(\eta, 1/\eta) \longrightarrow$ **Optimal CFL for $\eta^F \approx 1 \approx \eta^S$**
- $\eta^{f/s} \gg 1 \sim \mathcal{O}(1/h)$ -stabilization $\longrightarrow$ **degrades CFL**